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EDITED BY
Clint Bowers,
University of Central Florida, United States

REVIEWED BY
Triton Ong,
Doxy.me Inc., United States
Tiago Guerreiro,
University of Lisbon, Portugal

*CORRESPONDENCE
Joshua Juvrud,
Joshua.juvrud@speldesign.uu.se

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SafeSpaceVR: development and evaluation of an open-source, clinician-guided virtual reality tool for social anxiety

Kevin Hofbauer and Joshua Juvrud*

Department of Game Design, Uppsala universitet - Campus Gotland, Visby, Sweden

Background: Existing Virtual Reality Exposure Therapy (VRET) tools are predominantly solo-user platforms that lack real-time clinical interaction, creating structural and psychological barriers to the therapeutic alliance.
Objective: The aim of this study was to assess the feasibility and patient-side usability of SafeSpaceVR, an open-source, asymmetric VRET architecture designed to bridge the gap between patient immersion and real-time clinician control.
Methods: Utilizing a four-phase User-Centered Design (UCD) framework, we conducted formative research through a public survey (N = 33) and semi-structured clinician interviews (N = 3) to define technical requirements. These insights informed the development of a functional prototype, which was subsequently evaluated in a summative playtesting phase (N = 11) to assess usability, clarity, and perceived therapeutic relevance.
Results: Formative data emphasized the critical need for emotional doorways, such as a virtual clinician's office, and granular clinician-side control over social density. Summative testing revealed high usability and strong perceived relevance for social anxiety treatment.
Conclusions: This study demonstrates the initial feasibility of open-source, clinician-centered VR tools. As a proof of concept, SafeSpaceVR provides a foundational roadmap for remote, collaborative, and asymmetric digital mental health interventions.

KEYWORDS
anxiety disorders, CBT (cognitive behavioural therapy), clinical tools, exposure therapy, participatory design, virtual reality, VRET

Introduction

Virtual reality (VR) has gained increasing attention in clinical psychology as a tool for delivering evidence-based interventions, particularly in the treatment of anxiety and trauma-related disorders (Carl et al., 2019; Juvrud et al., 2018; Maples-Keller et al., 2017). Yet, existing VR therapy tools, such as *Psious*, *Oxford VR*, and *Bravemind*, are typically designed for solo use by patients, often in clinic-based settings with minimal interaction capabilities within VR (Freeman et al., 2017; Rizzo et al., 2012). These systems generally rely on pre-scripted exposure scenarios that patients navigate independently, while clinicians remain outside the virtual environment. As a result, current platforms rarely support real-time clinical interaction or online multi-user functionality, which limits opportunities for clinicians to guide, adapt, and scaffold the therapeutic experience as it unfolds.

Some recent advancements in clinician-controlled VRET, such as the work by Ferreira et al. (2024), have begun to explore real-time adaptation of virtual environments. This research demonstrated that allowing clinicians to dynamically adjust stressors, such as audience behavior or environmental triggers, can lead to more personalized and effective exposure outcomes. However, such systems have historically functioned as proprietary or experimental 'black-box' solutions, often requiring symmetric hardware setups where both the clinician and patient are immersed or co-located. Consequently, while the mechanics of real-time adaptation have been validated, the asymmetric technical requirements and structural accessibility of such tools remain largely unaddressed, leaving a gap between experimental efficacy and practical clinical deployment.

A lack of synchronous interaction is particularly problematic given that clinical responsiveness and the therapeutic alliance are consistently identified as key predictors of treatment outcomes across modalities (Flückiger et al., 2018). Clinicians play a central role in delivering structured, evidence-based interventions that facilitate psychological change (Kazdin, 2021). As demand increases for accessible and flexible therapeutic availability, particularly digital and remote options, there is a growing need for technologies that can extend the reach and adaptability of clinical practice (Andersson et al., 2019; Carl et al., 2019). The rise of online therapy platforms such as *BetterHelp* in the United States and United Kingdom, *Mindler* and *Kry* in Sweden, and the global shift toward remote therapy after the COVID-19 pandemic, have demonstrated that online therapy can be effective and widely adopted (Wright and Caudill, 2020; Anderson et al., 2013).

According to Ong et al. (2024), only 6.25% of clinical therapists in the US have used VR in treatment. This low adoption rate is not necessarily due to lack of interest. Indeed, over 80% of clinical therapists rated VR therapy as moderately to extremely useful. The challenge for implementation seems to be from a combination of structural and practical barriers. These barriers include a lack of training, equipment costs, limited reimbursement policies, and the absence of clinical therapist-centered VR tools (Meyerbröker, 2021; Chung et al., 2022). To date, no tools allow clinicians and clients to interact remotely in a shared VR space, representing a significant barrier to leveraging VR in contemporary clinical practice.

There is a critical theoretical need to address the asymmetric requirements of the clinical dyad. While patients require high levels of immersion to achieve the presence necessary for successful exposure, clinicians require high-utility control and observational clarity to safely manage the session. Immersing the clinician alongside the patient can often be counterproductive, creating a hardware barrier and limiting the clinician's ability to multitask or consult physical clinical notes. Therefore, a gap exists for VRET frameworks that support asymmetric interaction, providing the clinician with a specialized control interface on standard hardware while the patient remains in the immersive environment. Furthermore, addressing the systemic black-box nature of proprietary tools via open-source, extensible architectures is essential for the long-term democratization of VRET, allowing clinical settings to bypass high procurement costs and adapt tools to their specific local workflows.

Addressing this gap, the present project utilized a User-Centered Design (UCD) process across several phases to develop to develop *SafeSpaceVR*, an open-source VR application. Designed as a small-scale proof of concept, *SafeSpaceVR* was developed through formative research by involving clinical therapists directly in the design process. The goal was to create a tool that integrates seamlessly into therapeutic workflows, enabling multi-user exposure scenarios guided by clinicians in real time. The present research emphasizes the feasibility and patient-side usability of this architecture though a summative evaluation, as a foundation for future clinical applications. It is important to note that the tool itself is not a treatment, nor a stand-alone therapeutic solution. Rather, it was designed as a supportive clinical tool in collaboration with clinicians, for use in therapeutic settings, both in-person and online. To this end, three research questions guided the project. First, we examined which design qualities are most critical for developing a VR exposure therapy tool that aligns with the practical needs of clinicians and patients in the treatment of social anxiety disorder (RQ1). Second, we explored which specific social situations clinicians and patients consider most relevant for practice within VR exposure therapy (RQ2). Finally, we assessed how users experienced the prototype in terms of usability, clarity, and perceived usefulness for therapeutic purposes (RQ3).

Social anxiety disorder

Virtual reality (VR) shows particular promise as a tool for clinicians treating Social Anxiety Disorder (SAD), one of the most prevalent anxiety disorders, with lifetime prevalence rates reaching up to 12% (Kessler et al., 2005). Cognitive Behavioral Therapy (CBT), and specifically exposure therapy, is considered a first-line treatment for SAD due to its robust evidence base and demonstrated long-term efficacy (National Institute of Mental Health, 2001). Conventional, in-person exposure therapy, however, presents practical challenges; these include difficulties in staging or accessing appropriate social situations, risks to patient confidentiality, uncontrollable environmental variables, and additional costs related to travel or logistics (Hawajri et al., 2023).

Exposure therapy is a behavioral treatment strategy that aims to reduce fear and avoidance by gradually and repeatedly exposing individuals to anxiety-provoking situations in a controlled and safe environment. Repeated exposure helps individuals re-evaluate their feared stimuli and form new, non-threatening associations (Craske et al., 2014). Over time, this process leads to habituation and reduced anxiety responses. In the context of SAD, exposure therapy often involves confronting anxiety provoking social situations such as public speaking, initiating conversations, or being observed by others (Heimberg et al., 2010). VRET builds on this model by simulating these situations digitally, allowing for more control, repeatability, and customization (Parsons and Rizzo, 2007).

While traditional remote interventions via 2D video conferencing (e.g., Zoom) can facilitate talk therapy, they often fail to trigger the physiological and psychological "sense of presence" necessary for effective exposure (Wiederhold, 2020). In the treatment of SAD, immersion is a critical mechanism; it creates a "perceptual illusion" of being observed by a real audience, which is required to activate the fear structures that clinicians aim to modify. VR provides a level of sensory fidelity and spatial embodiment that 2D platforms cannot, making it uniquely suited for simulating the complex social dynamics, such as eye contact and physical proximity, that are central to social anxiety triggers.

VR provides a controlled, repeatable, and safe environment in which clients can practice anxiety-provoking scenarios (such as public speaking, group interactions, or other social situations) and potentially lowering barriers to treatment and increasing accessibility (Parsons and Rizzo, 2007). Unlike traditional exposure therapy, which requires clinician supervision in real-world environments, VR has the potential to replicate these social triggers within a virtual context, facilitating practice even when in-person interactions are limited (Andersson, 2015).

Despite these advantages, existing VR tools for exposure therapy are largely designed for solo use and lack support for real-time clinician interaction (Meyerbröker, 2021). They are rarely integrated into clinical workflows, limiting their practical application in therapy even when there is high interest from clients. Addressing these limitations by designing VR systems that support clinician-guided, interactive, and contextually relevant exposure represents a critical step toward enhancing the accessibility and effectiveness of exposure-based treatments, such as for SAD.

Designing a methodology for Virtual Reality Exposure Therapy (VRET)

Virtual Reality Exposure Therapy (VRET) has received increasing attention as a modern alternative to traditional real-world exposure therapy, particularly for treating anxiety disorders. In a systematic review of six randomized controlled trials, Chowdhury and Khandoker (2023) concluded that VRET is as effective as real-world exposure in reducing social anxiety symptoms. Despite this evidence of efficacy, adoption of VRET in clinical practice remains limited. One contributing factor is the challenge of designing VR tools that are both clinically effective and practically usable within therapy workflows.

Designing effective VR-based exposure tools requires careful attention to the balance between immersion and psychological safety. Key success factors in exposure therapy include a sense of control, predictability, and gradual progression (Anderson et al., 2013). Research in VR contexts reinforces these principles. For example, Herrero et al. (2014) found that customizable settings and real-time clinical support reduce dropouts and enhance treatment efficacy, while Powers and Emmelkamp (2007) demonstrated that progressive exposure, i.e., beginning with low-intensity scenarios, is more effective than immediate immersion in high-anxiety situations. These insights should directly inform the design of a VR tool for clinicians, one that incorporates adjustable environment settings, audience sizes, and presentation formats to optimize both safety and therapeutic effectiveness.

Furthermore, key features known to enhance exposure therapy outcomes, such as predictability, perceived control, and clinician presence, should also be integrated into the VR tools (Anderson et al., 2013). This ensures that users are not overwhelmed, and that each session can be adjusted in real-time, either during in-person sessions or online *via* remote access. For example, features that allow clinicians to gradually increase exposure levels (e.g., audience size or engagement level) and to create a safe progression of scenarios tailored to individual needs (Powers and Emmelkamp, 2007).

Designing believable yet psychologically safe social environments is also crucial. The National Institute of Mental Health (2001) highlights that common social anxiety triggers include public speaking, talking to authority figures, and group interactions. These settings area situational and require not just visual believability, but also contextual interactivity to feel authentic. To support user needs, VR tools should allow for scene customization based on these common triggers, aligning with clinician feedback and academic guidance on anxiety-specific scenario design.

Current study

While existing literature supports the feasibility and promise of VR exposure therapy, it also highlights major gaps in current implementations. There is a lack of clinician-controlled, multi-user VR tools specifically designed for remote use that allow personalized customization by the clinician for a patient's individual needs. This project addresses these gaps by developing a clinician-guided multi-user VR tool, *SafeSpaceVR*, that allows clinicians and patients to interact in the same virtual space. The tool was designed to include scenario customization, clinician-controlled exposure levels, and collaborative elements like shared PowerPoint slides that include text and images as a presentation tool. It is important to note that certain challenges still remain in the current study, such as limited patient access to VR hardware and clinician training. To best address these, the VR tool was designed to prioritizes accessibility, clinician input, and modular design, laying the groundwork for more adaptable and widely usable digital therapy tools.

Player-centered design and emotional safety

From a design perspective, this project applied principles from a player-centered design and emotional safety in interactive systems. The *SafeSpaceVR* tool emphasizes usability, accessibility, and psychological safety throughout the experience. Designing for clinical use requires sensitivity to user vulnerability. This includes giving users agency to control their exposure (e.g., starting in a familiar environment), providing clear transitions (e.g., an onboarding space in a neutral environment, such as in a virtual clinician's office), and avoiding overstimulation or sudden changes that may trigger anxiety.

Moreover, the clinician's ability to adjust exposure levels to meet individual patient's needs supports a responsive and adaptive experience that not only mirrors traditional therapy but makes it more flexible. These design choices follow Herrero et al.'s (2014) guidance on creating emotionally secure virtual environments for therapy.

Finally, the inclusion of real-time clinician presence within the VR world draws from both exposure therapy's emphasis on guided treatment and multi-user design principles that emphasize social presence and co-regulation in shared virtual spaces (Geraets et al., 2019). By allowing clinicians to join the virtual world, either to participate or observe, the tool supports trust, customization, and feedback, with are essential elements for successful therapeutic engagement.

Design protocol

The present study focused on Social Anxiety Disorder (SAD), a prevalent condition for which exposure-based interventions are

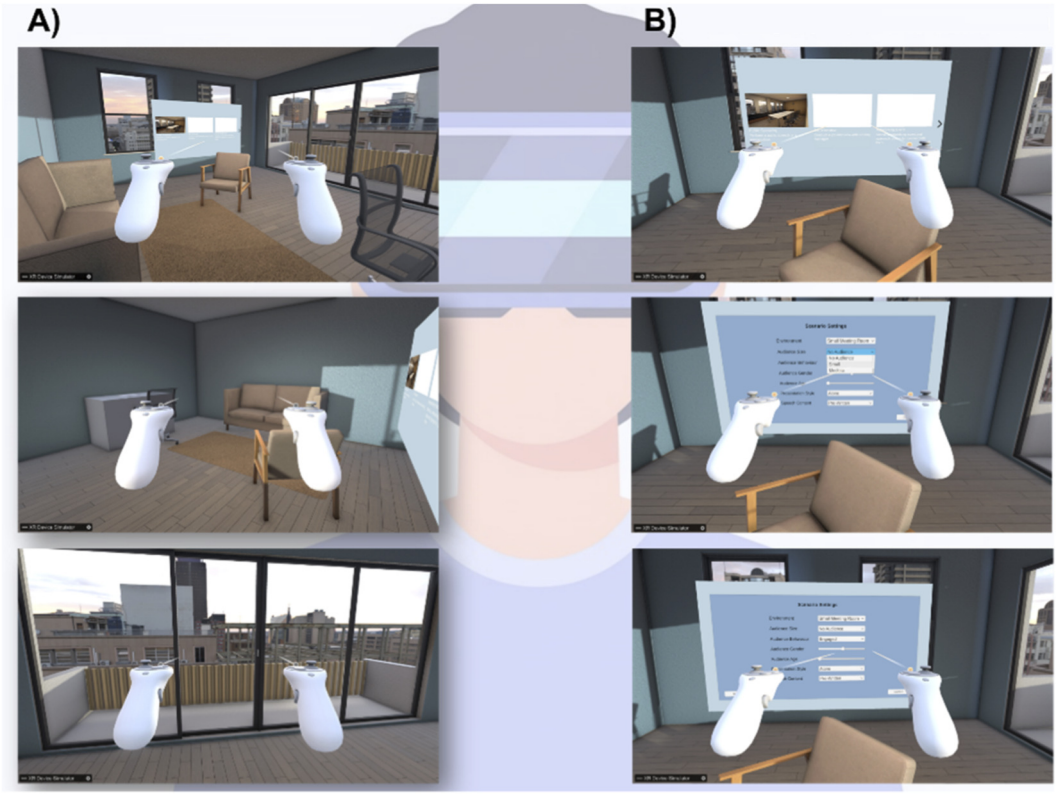


FIGURE 1
This figure illustrates two key aspects of the VR experience in *SafeSpaceVR*. **(A)** showcases sample images of the immersive VR clinical office, demonstrating the virtual setting where the patient and clinician launch into the application. **(B)** presents sample images of the scenario selection menu, highlighting the user interface for choosing different therapeutic or training scenarios within the VR application.

considered a first-line treatment. SAD provides a well-defined context in which the limitations of traditional real-world exposure therapy are pronounced, making it an ideal target for testing the potential of VR-based interventions. We adopted an multi-phase User-Centered Design (UCD) framework. In the formative phases (Phases 1 and 2), we gathered requirements from end-users and experts. In the development and evaluative phases (Phases 3 and 4), we translated these requirements into a functional prototype and conducted a usability assessment.

In the first phase, a survey collected quantitative and qualitative data from individuals to identify common social anxiety triggers, assess familiarity with VR, and gather preliminary preferences for a therapeutic VR tool. Insights from this survey informed the second phase, semi-structured interviews with licensed clinicians, which sought expert perspectives on clinical needs, workflow integration, and potential barriers to VR adoption in therapy. The third phase, tool development, translated the combined findings from survey and interviews into a functional VR artifact that incorporated both user preferences and clinical requirements. Finally, playtesting evaluated the tool's usability, clarity, and perceived therapeutic usefulness, providing direct feedback to assess the effectiveness of design decisions and inform future refinements. This UCD approach allowed the study to systematically combine user-centered insights with clinical expertise, ensuring that the resulting *SafeSpaceVR* tool is both practically and therapeutically relevant.

Materials and equipment

The developed tool is a multi-user VR application designed to support clinicians in guiding clients through customizable social scenarios. It was developed with the HTC Vive headset as the primary target hardware, using Unity's XR Interaction Toolkit to ensure compatibility with VR input systems. SteamVR was used as the runtime platform to establish communication between the headset and the application, enabling stable integration of VR hardware into the Unity-based prototype. The tool allows two users to interact in the same virtual space: one in VR (the client) using a headset, and the other in first-person mode on a standard computer (the clinician). The final application was compiled as a standalone Windows executable (.exe) to ensure ease of deployment for clinicians. To facilitate review and further testing, the. exe build of the prototype, can be downloaded from: https://drive.google.com/file/d/1vatjhmqLFYwurfG1NiEUrNYuVTSrsVDX/view?usp=sharing.

Upon launching the application, users start in a 3D-modeled clinical office (Figure 1), which functions as a main menu and onboarding space. This neutral environment helps orient the client and allows time for adaptation before entering anxiety-inducing

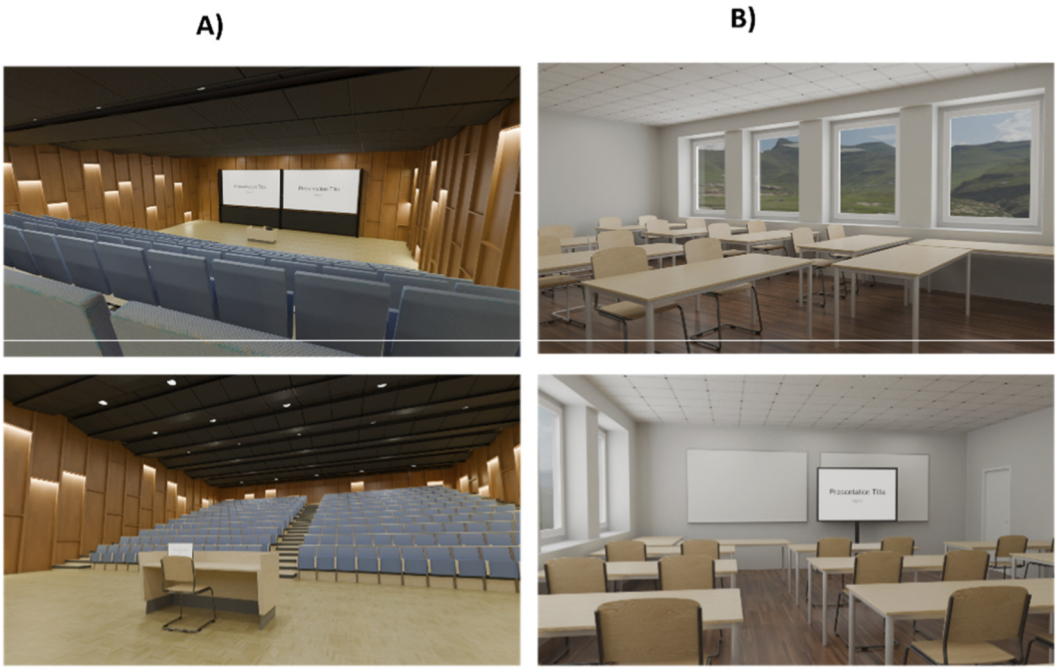


FIGURE 2
This figure demonstrates two environments in *SafeSpaceVR*: **(A)** a lecture hall, and **(B)** a classroom.

situations. The decision for this transitional space was informed by clinician interviews, where it was described as a way to create emotional safety at the beginning of a session (see Interview Analysis). Once inside the virtual office, the clinician and the client can choose from a variety of scenarios. During testing, the implemented scenario was the public speaking exercise, where the clinician can adjust personalized settings for treatment.

Virtual environment

Users can choose between four different environments: a conference room, classroom, lecture hall, and auditorium (see Figure 2 for lecture hall and classroom examples). The conference room offers an intimate space with seating for up to nine people, including the clinician. The low ceiling and small size of the room offers a comfortable starting place for clients who are just beginning exposure exercises. The classroom setting expands capacity to around 35 seats, with bright lighting and fixed desks. This environment allows clients who are still in school or work spaces with similar environments to find familiarity to life situations. For larger-scale practice, the lecture hall holds roughly 100 attendees, featuring elevated seating, a raised podium, and projection screens. This is useful for clients targeting large university, conference, or public speaking presentations. Finally, the auditorium accommodates up to 300 virtual participants in a spacious, is a dimly lit venue with professional stage lighting and a balcony level. This simulates large-scale speaking engagements such as professional talks or public ceremonies. Selecting the appropriate environment allows clinicians to tailor personalized scale and atmosphere to each client's comfort level, therapeutic goals, and current life situation.



FIGURE 3
This figure gives an example of one of the available models in *SafeSpaceVR*.

Virtual audience

Within each environment, audience size can be adjusted in four increments: no audience (0% of the seats filled), small (roughly 20%), medium (50%), and large (100%). Adjusting the number of avatars present directly influences the previewed social density, enabling clinicians to introduce clients gradually to larger crowds. For example, a client may begin with an empty room to focus on voice projection and presentation timing, then progress to a

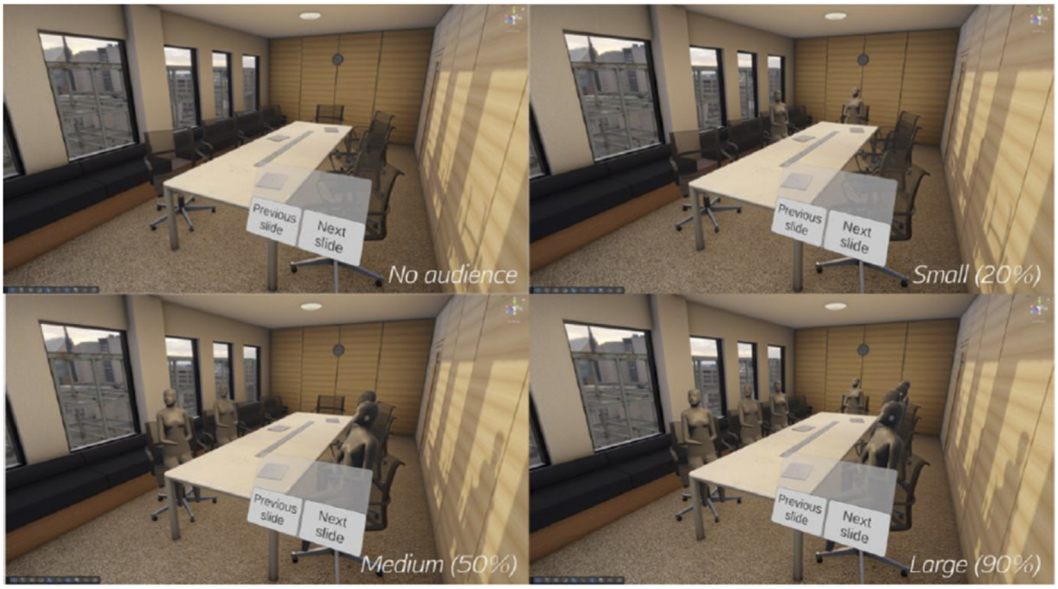


FIGURE 4
This figure shows various audience sizes in the conference room environment (with placeholder character models).

small group where perceived interpersonal observation emerges, before eventually confronting a full room to practice maintaining composure under maximal social pressure. This feature of gradual exposure is informed by both existing research (e.g., Smith, 2025) and results from clinician interviews. See Figure 3 for an example of an audience character model and Figure 4 for examples of audience sizes.

Virtual audience behavior

The tool offers four behavioral profiles for the virtual audience: neutral, engaged, distracted, and challenging. Each is designed to provoke different aspects of public-speaking anxiety. In neutral mode, avatars sit attentively, remaining largely expressionless to create a baseline speaking experienced. The engaged profile introduces subtle positive feedback (e.g., nodding, smiling, and note-taking) to help clients build confidence. A distracted audience occasionally looks at their phones, whispers to neighbors, shifts restlessly, or makes interrupting noises, encouraging clients to maintain focus despite minor disruptions. Finally, the challenging setting features avatars who frown, fold their arms, or ask pointed questions, simulating an impatient audience that targets more advanced exposure goals. This feature was informed by the results of both survey respondents (see Survey Analysis) and by clinician interviews (see Interview Analysis).

Virtual audience representation

In addition to size and behavior, clinicians can tailor the demographic of the virtual audience by selecting the age profile: younger adults (18–35), older adults (50+), or a balanced mix. Younger audiences are visually distinct, featuring casual clothing that evokes informal gatherings. Audience wears casual clothing like t-shirts, hoodies, and jeans, and are arranged in a less formal seating plan, simulating environments like a university classroom or a local meetup. Older audiences appear in more formal clothing in traditional seating, simulating professional gatherings or conferences. A mixed demographic blend both styles together. In addition to age groups, clinicians can also adjust gender of the audience. One clinician noted in interviews that this feature helps target exposure for clients who only feel anxious when interacting with one specific gender (see Interview Analysis).

Virtual presentation style

To mirror the variety of public-speaking contexts, the clinician can choose from four distinct presentation modes: solo, co-presentation, practice, and question and answers (Q&A). The solo mode is designed for the client to speak alone on stage, which teaches them to manage their anxiety without any external support. For clients needing more guidance, the co-presentation option allows the clinician to join them on stage, providing real-time support and helping to build confidence by being directly involved in the session. Practice mode offers a relaxed, pressure-free setting where the client can rehearse their content at their own pace, with the ability to pause or restart as needed without any audience pressure. This is an ideal way for a client to build familiarity with the virtual environment before engaging in a more active scenario.

The Q&A mode allows clients to engage directly with the virtual audience by being presented with the opportunity to respond to questions. The tool allows clinicians to manage these questions in two distinct ways to mirror real-world unpredictability, while maintaining a controlled therapeutic environment. First, the system includes a library of pre-loaded questions based on common public speaking fears, which provides a quick, ready-to-use option. Alternatively, and more critically, the clinician can customize questions in real-time by either typing or speaking

them into the system. This allows for a highly personalized experience where the clinician can target a client's specific anxieties and practice scenarios with questions directly relevant to their real-life situations. This exercise boosts adaptability and reduces the fear of the unknown.

Virtual speech content

The tool supports flexible content delivery by offering two speech modes: pre-written and improvisation. In pre-written mode, clients can choose from several built-in speeches or custom speeches. Custom speeches are uploaded by the client or clinician using a PDF-file. They then deliver the prepared script while being observed by the clinician. Alternatively, in improvisation mode, clients come up with a speech for a randomly selected topic on the spot, with or without the help of presentation slides. This simulates unpredictable real-world speaking demands and reinforces adaptive thinking under pressure.

Summary

The designed settings allow clinicians to tailor the experience based on a client's therapeutic goals, individual needs, and level of comfort. Audience size and reactions are critical variables in public speaking anxiety (Anderson et al., 2013), and adjustable settings support gradual exposure, which is a key principle of successful treatment (Chowdhury and Khandoker, 2023).

Methods

The *SafeSpaceVR* tool developed in this study is not merely a product, but a central investigative tool used to generate knowledge and answer the research questions of the study. The overall research design was structured to ensure that each phase of data collection and analysis directly informed the development of the VR tool. This involved a continuous cycle of planning, analysis, implementation, and evaluation, with each cycle building upon the previous one (Gould and Lewis, 1985). The specific methods, surveys, interviews, and playtesting, were strategically integrated into the design process to gather diverse perspectives and validate design decisions. This loop can be visualized as.

1. Phase 1: Formative User Research (public survey): This initial quantitative and qualitative data collection aimed to understand the broader user landscape, identify common social anxiety triggers, assess VR familiarity, and gather initial preferences for a therapeutic VR tool. This provided foundational user-centered data to begin conceptualizing the prototype.
2. Phase 2: Expert Requirement Gathering (clinician interviews): Building on the survey, these semi-structured interviews sought in-depth, expert perspectives from licensed clinical therapists. The goal was to understand clinical needs, practical considerations, and potential barriers to integrating VR into therapy workflows, ensuring the tool's clinical relevance and usability from a professional standpoint.
3. Phase 3: Prototyping and Technical Implementation (final tool development): This central phase involved translating the combined results from the survey and clinician interviews into a functional VR artifact. It was the tangible manifestation of the research, incorporating user preferences and clinical requirements into a final testable tool.
4. Phase 4: Summative Usability Testing (playtesting): This final evaluation phase provided direct user feedback on the developed prototype's usability, clarity, and perceived therapeutic usefulness. It assessed how well the design decisions from previous phases translated into a positive user experience and to inform future refinements.

This approach ensured that the VR tool was not developed in isolation but was continuously shaped by empirical data and expert feedback. This methodology directly addresses the research questions: RQ1 (which design qualities are most critical for developing a VR exposure therapy tool that aligns with the practical needs of clinicians and patients in the treatment of social anxiety disorder) was addressed through the UCD framework, informed by user and clinician needs. RQ2 (which specific social situations clinicians and patients consider most relevant for practice within VR exposure therapy) was answered through the survey and interview phases, directly influencing scenario selection. RQ3 (how users experienced the prototype in terms of usability, clarity, and perceived usefulness for therapeutic purposes) was evaluated through the playtesting. The methods and protocols of each phase of the study were conducted in accordance with the standards specified in the 1964 Declaration of Helsinki and approved by the local ethics committee, the Swedish Ethical Review Authority (2023-04552-02).

Phase 1: survey

A survey was used to gather broad user perspectives on social anxiety, VR familiarity, and initial preferences for a therapeutic VR tool. This stage was crucial for understanding the general landscape of user needs and informing the conceptualization and early prototyping stages of the VR application, particularly in identifying initial feature preferences (RQ1) and relevant social situations for exposure therapy (RQ2).

Participants

Participants were recruited online via social media, online forums, and posted flyers on a university campus. This recruitment strategy was chosen for its effectiveness in reaching a diverse and relevant population interested in VR and mental health, given the project's scope and resources (Arigo et al., 2018). A total of 33 respondents completed the survey (target sample size 30, aimed at gathering a broad cross-section of social anxiety triggers to inform the formative design). The sample was diverse in age, ranging from 18–55 years (M = 26 years, SD = 8.1 year), with the majority of respondents (57.6%) being 18–24 years old. A significant portion (54.5%) stated that they have attended or are currently attending therapy or counselling, while 9.1% preferred not to answer this question. No prior VR experience was required to participate, which

was a deliberate choice to understand the initial perceptions of a broad user base, including those new to immersive environments.

Materials

The survey was conducted using Google Forms. The choice of Google Forms facilitated broad participation and streamlined data collection. The survey was structured into four thematic sections to systematically gather relevant information: social anxiety and related experiences, digital tool and VR usage, preferences for VR in therapy, and demographic information.

The survey had a total of 20 questions structured across four main sections. The first, on social anxiety and related experiences, contained a mix of Likert-scale questions, such as "How comfortable do you feel in the following social situations?", and a multiple-choice question. The second section, on digital tool and VR usage, consisted of three Likert-scale questions and one open-ended question to gauge familiarity with relevant technologies, for example, "Have you ever tried Virtual Reality (VR) before?". The third section, focusing on preferences for a VR therapy tool, was comprised of four Likert-scale questions and five open-ended questions designed to directly inform the prototype's features, including a question asking, "If you were to use a VR tool to practice social situations in a therapy setting, what features would be important to you?". Finally, the demographics section included two Likert-scale questions, a yes/no question about interest in future playtesting, and an optional open-ended question for contact details. The survey took participants approximately 10–15 min to complete.

Data analysis

Data collection was open for 2 weeks, after which the survey was closed. All responses were reviewed by a researcher and checked for completeness. To maintain the quality of the dataset, any duplicate or incomplete entries were removed. The analysis consisted of two main components: quantitative analysis of the closed-ended questions and qualitative content analysis of the open-ended responses.

For the quantitative data, responses to multiple-choice and Likert-scale questions were compiled and scored. Each item was first categorized according to its topic, such as comfort with social situations or willingness to use VR in therapy. Mean scores were calculated for Likert-scale questions to help identify trends across the dataset, such as which social situations participants found most uncomfortable. These numerical summaries helped determine which features and scenarios were most relevant for the design of the VR tool. The results were then visualized in the form of bar graphs and charts to facilitate interpretation and comparison.

In the case of open-ended questions, a qualitative content analysis was conducted to systematically identify, categorize, and interpret themes within the textual data. This process followed the guidance of established approaches outlined by Elo and Kyngäs (2008), Erlingsson and Brysiewicz (2017), and Luo (2023).

The first step to the qualitative analysis was familiarization. All open-text responses were compiled into a single document and reviewed by the researchers to develop an overall sense of the content and to become familiar with the data. This initial reading helped in understand the breadth and depth of participant feedback. Any irrelevant or placeholder responses, such as "N/A" or "-", were excluded at this stage.

The second step was the initial coding (open coding) of responses. Meaningful phrases, sentences, or ideas that appeared significant or recurrent were highlighted. These segments were then labelled with initial descriptive codes that captured their essence (e.g., "realism," "pause button," "therapist support"). This was an inductive process, allowing codes to emerge directly from the data.

Next, the researchers categorized the codes. The initial codes were grouped into broader categories reflecting thematic concerns. This involved clustering similar codes together to form overarching themes, such as "desired features," "emotional safety," or "reasons for avoidance". This step organized the raw data into a more structured format.

Once grouped, the categories were reviewed and refined to ensure consistency and to confirm that each coded item clearly fit the intended theme. This process involved re-reading the data with the emerging categories in mind, adjusting definitions, and merging or splitting categories as necessary to achieve conceptual clarity.

The researchers then used frequency counts to track how often certain themes appeared, providing an indication of their prevalence and importance among participants. Additionally, representative quotes were selected to illustrate key findings and provide direct participant voices to support the identified themes. The coded results were organized into tables, which were used both for reporting and to guide design decisions.

The last step in the qualitative analysis was a process of triangulation to strengthen the validity of the findings. This involved comparing and cross-referencing patterns from the survey data with insights from the literature review (and later, again with the clinician interviews). For example, participant interest in audience customization and scene control was not only a repeated theme in open-ended responses but also aligned with feedback from clinicians who emphasized the need for realism and clinician-guided control in exposure therapy. Similarly, user concerns about emotional safety and needing the ability to exit a scenario were mirrored by both published literature (Anderson et al., 2013; Herrero et al., 2014) and the professional insights gathered during interviews. This comparative process helped ensure that the conclusions drawn from the survey were consistent with existing knowledge and grounded in multiple sources of evidence, thereby informing the design process with robust, multi-faceted data.

Results and design implications

Social anxiety triggers

Participants rated talking to strangers (mean comfort rating M = 2.9, SD = 1.2), making phone calls (M = 3.1, SD = 1.2), and public speaking (M = 3.1, SD = 1.1) as the most anxiety-provoking contexts. This, together with the fact that 66.7% of participants would like to practice public speaking, informed the design decision to prioritize a public speaking scenario for the tool.

Avoidance behavior

A substantial 75.8% of participants reported sometimes or often avoiding social situations due to discomfort or anxiety. The

most common reasons cited for this avoidance were "worrying about what to say" (54.55%), "fear of making mistakes" (54.55%), and "fear of embarrassment or judgement" (48.48%). These findings underscored the critical need for a safe, repeatable exposure environment that could address these core anxieties.

Desired features for a VR therapy tool

"Realism and immersion" emerged as the most frequent request (mentioned 13 times), with several respondents stressing the importance of believable environments and character behavior for provoking authentic responses. Respondents also reported highly valuing user agency, specifically the ability to pause or exit the experience at any time (5 mentions), and scenario variety (5 mentions), such as different environments and audience types. For emotional safety, control over the scenario (e.g., ability to exit if overwhelmed) was paramount (8 mentions), along with the importance of direct clinician involvement (6 mentions). The strong desire for "realism and immersion" and user agency (e.g., pause/exit options) informed the tool's visual design and interaction features, ensuring alignment with user expectations for an effective therapeutic tool.

Openness to VR therapy

A large majority (78.8%) of respondents indicated openness to using VR with a therapist to practice social situations, suggesting strong potential for adoption due to its usefulness as a starting point before real-world exposure and the privacy and convenience it affords.

VR and digital mental health tool usage

A notable observation was the high reported prior VR usage for gaming (78.7%) contrasted with a significantly lower engagement with digital mental health tools (27.3%). This pattern suggested a significant opportunity: users are already comfortable and familiar with VR as a technological medium, but there appears to be a lack of appropriate, clinically relevant VR applications for therapeutic purposes. This implied that the barrier to adoption for therapeutic VR might not be the technology itself, but rather the availability and design of specialized tools. By leveraging existing VR familiarity for a novel therapeutic context, the VR tool directly addresses this gap, aiming to normalize VR for mental health applications and potentially broaden its reach. This also indicated that the design could draw upon familiar game design principles to ease user onboarding and interaction.

Remote therapy acceptance

Responses to whether participants would be more likely to try exposure therapy if it were offered remotely were mixed: 36.4% said yes, 21.2% said no, and 42.4% were unsure. This suggests that while remote delivery may enhance accessibility for some, others remain cautious or undecided, highlighting the need for flexible delivery models and clear communication about the benefits of VR therapy.

Phase 2: clinician interviews

Semi-structured interviews with licensed clinicians constituted a critical qualitative data collection phase within the design process. These interviews were conducted after the initial survey to gain in-depth, expert perspectives that were essential for understanding the clinical context, identifying practical needs, and informing the design of a therapeutically relevant VR tool (RQ1). This phase deepened the understanding of clinician workflows, their specific needs, and potential barriers to integrating VR into their practice, going beyond the general user preferences identified in the survey.

Participants

Three licensed clinical therapists, including one female and two males, participated in the interview portion of this study. A sample of three expert practitioners was deemed sufficient to reach thematic saturation regarding high-level clinical workflow requirements, consistent with qualitative research standards for specialized expert populations (Hennink and Kaiser, 2022). They were selected through snowball sampling, with recruitment focused on practitioners in Sweden and the United States. Clinical therapists who agreed to participate were sent a detailed information sheet and asked to sign a digital consent form before the interview. All interviews were conducted in English via Microsoft Teams. All three clinical therapists were given an informed consent form before participation and were informed of their right to withdraw at any point without consequence. Their diverse perspectives helped shape both the design of the VR prototype and the interpretation of findings. Demographic details such as age were not collected systematically, as the focus of the interviews was on professional experience and therapeutic context relevant to the study's aims.

Clinical therapist 1 (T1): A licensed psychodynamic therapist based in California, United States, with 11 years of clinical practice experience. T1 worked full-time in private practice, seeing 20–22 clients per week, entirely through online sessions for the past 5 years. Their primary areas of focus included emotional regulation, anxiety disorders, and self-esteem issues. Although not currently offering exposure therapy, they expressed strong interest in new technologies for clients facing barriers to in-person interaction.

Clinical therapist 2 (T2): A licensed psychologist based in Sweden, licensed in September 2024, following a year of practical training in cognitive behavioral therapy (CBT). T2 works for a private mental health company offering digital therapy *via* its own platform, managing approximately 15 clients per week. Their work focuses on CBT-informed treatment plans, emphasizing anxiety, avoidance behavior, and exposure-based techniques. T2 noted that digital tools are already embedded in their workflow but highlighted the need for more interactive and immersive solutions.

Clinical therapist 3 (T3): A licensed clinical therapist based in California, United States, licensed in 2019 after beginning training in 2012. T3 worked as an independent practitioner, seeing 15–18 clients per week, with roughly half of the sessions taking place online. Their therapeutic orientation integrates humanistic and behavioral approaches, with expertise in trauma processing, social anxiety, and life transitions. T3 regularly uses gradual

exposure techniques and expressed enthusiasm for tools that could enhance remote therapy by improving engagement and observational detail.

Materials

A semi-structured interview guide was developed specifically for this study, consisting of 15 open-ended questions grouped into six thematic areas: (1) clinician background, (2) use of digital tools in therapy, (3) exposure therapy practices, (4) social anxiety triggers and therapy structure, (5) opinions on desired features and the future of VR therapy, and (6) cultural and accessibility considerations. The aim was to elicit rich qualitative data about clinical therapists' clinical routines, attitudes toward VR, and design-relevant insights that could inform prototype development. The semi-structured nature allowed for flexibility, enabling the interviewer to pursue interesting points raised by the clinicians while ensuring all key topics were covered.

All interviews were conducted via Microsoft Teams with the participants' consent. At the end of each session, participants were offered a short demonstration of the VR prototype, followed by an informal discussion in which they were invited to share immediate feedback on its clinical relevance, design, and potential use cases. This demonstration provided a concrete reference point for therapists to offer more specific and actionable feedback.

Procedure

To ensure participant comfort and open discussion, interviews were conducted in a conversational tone, following the semi-structured guide while allowing for natural follow-up questions. The researcher took live notes during the interviews and immediately reviewed for completeness after each session. This approach allowed for accurate content analysis while maintaining participant privacy and data security. Each interview session lasted approximately 30–45 min.

Data analysis

After conducting the interviews, the interview notes were cleaned to correct any spelling or grammatical errors. Following this, a qualitative content analysis was performed, drawing on established approaches outlined by Elo and Kyngäs (2008), Erlingsson and Brysiewicz (2017), and Luo (2023). This systematic process ensured a structured and transparent analysis of the qualitative data.

The first stage involved familiarization with the data. All interview notes were read multiple times to develop an overall sense of the content and to become immersed in the data. This reading helped in identifying initial patterns and gaining a holistic understanding of the clinicians' perspectives.

During the next stage, open coding was conducted. This involved highlighting specific phrases and ideas that appeared meaningful or recurrent across the interview notes. These segments were then labelled with initial descriptive codes that captured their essence, such as "realism importance," "cost barrier," "therapist control," or "onboarding environment". This inductive coding process allowed themes to emerge directly from the data without imposing preconceived categories.

Following open coding, the identified codes were grouped into broader, overarching categories to begin forming a thematic structure. Categories such as "desired features," "adoption barriers," "current practice," and "conditions for effective use" were developed by clustering similar codes together. This step involved a constant comparative method, where new codes were compared to existing categories to refine their definitions and relationships.

Once grouped, these categories were reviewed and refined to ensure consistency and to confirm that each coded item clearly fit the intended theme. This process involved re-evaluating the coherence of categories and themes, merging redundant ones, or splitting overly broad ones to achieve greater precision and clarity.

The final step involved quantifying the frequency of each category to assess its relative importance across all three interviews. This provided a quantitative overview of the qualitative data, highlighting dominant themes. Additionally, illustrative quotes were selected from the interview notes to provide direct evidence and rich context for each theme. This process allowed for the identification of dominant themes and their alignment with the project's research questions. These themes were used to inform the design decisions of the *SafeSpaceVR* tool and were cross-referenced with survey findings to enable triangulation of insights. All coding and analysis were conducted using a structured spreadsheet.

Results and design implications

Current exposure therapy practice

Clinicians consistently reported using structured, goal-oriented techniques that progress gradually, primarily tracking patient progress through client self-reports and their own clinical notes. This highlighted a need for a VR tool that could integrate seamlessly with existing workflows and potentially offer more objective metrics to supplement subjective assessments, thereby enhancing clinical efficiency and data collection.

Desired features in a Vr exposure therapy tool

A dominant theme across all interviews was the paramount need for realism and customizability in a VR exposure therapy tool. Clinicians stressed that virtual environments must closely mimic real-world social situations to provoke genuine emotional responses. T2 stated, "The tool needs to feel real enough that it triggers the right parts of the brain, like a real situation would." Realism, according to participants, includes lifelike virtual characters, natural social dynamics, and responsive audience behaviors.

Customization was also deemed essential

T3 noted, "It would be great to adjust the age, gender, or engagement level of the audience depending on the client's specific fears." Similarly, T2 said, "Having the option to change the size of the audience or whether they're looking at the client is very important." These preferences demonstrate that clinicians want granular control over environmental variables to match the structure of graded exposure therapy.

Clinicians also highlighted the importance of clinician control and shared agency. While they acknowledged that client preferences

matter, clinicians ultimately wanted to guide the process. T1 said, it "should be a conversation between patient and clinician, but ultimately the clinician should decide."

Several clinicians also mentioned interest in integrating biometric data to support assessment. T3 said, "Physiological measures would be amazing, such as heart rate, heart rate variability, skin response, and possibly even EEG." This quote reflects a desire to incorporate objective feedback into sessions, as a way to supplement the self-reports that currently guide treatment.

Design implications

The consistent need for realism and customizability led to the development of photorealistic 3D environments, customizable audience sizes, and planned features for audience behaviors and demographics. The emphasis on clinician control and shared agency influenced the multi-user functionality, allowing clinicians to guide and observe the client's experience dynamically. The desire for objective feedback prompted the inclusion of API placeholders for biometric data integration. The clinicians' emphasis on "realism" extending beyond visual fidelity to include "natural social dynamics" and "responsive audience behaviors" is particularly significant. This implies that simply making environments look real, as achieved with photorealistic 3D assets, is insufficient for true therapeutic efficacy. Effective exposure therapy requires believable social dynamics, where the virtual audience's behavior and reactions feel authentic and responsive to the client's actions (Parsons and Rizzo, 2007).

The desire for social interaction elevates the design challenge from primarily graphical fidelity to complex behavioral simulation, directly influencing the "Audience Spawn System" and its planned future features, which aim to incorporate configurable audience behaviors like engaged, distracted, or challenging responses. This understanding guides the development towards creating truly interactive and emotionally impactful scenarios. A successful therapeutic tool must balance the clinician's need to guide the exposure process with the client's psychological need for safety and autonomy. This implies that the design of the user interface and interaction flow should offer clear clinician control while simultaneously providing client-side affordances, such as pause options. This balance is critical for therapeutic efficacy, as forcing an experience can be counterproductive to building resilience.

Barriers to adoption and implementation

Several barriers were identified, including the cost and complexity of VR equipment, a perceived lack of training among clinicians, concerns about data privacy and the need for robust security protocols, and issues with insurance reimbursement. These practical constraints were critical in shaping the prototype's design and future development considerations, leading to design choices such as desktop fallback options, making sure no data of any kind was stored online, and a focus on user-friendly setup to mitigate these concerns.

Conditions for effective use and implementation

Clinicians emphasized that VR exposure should serve as a supplementary tool, augmenting clinician-led care rather than replacing it. The tool was seen as particularly helpful for clients who lack natural opportunities for exposure, such as those not currently engaged in work or school environments. The inclusion of a neutral, onboarding environment, such as the virtual clinician's office, was highly valued, described as a "virtual doorway" before entering more stressful scenarios. The clinicians' support for a neutral onboarding environment directly resulted in the creation of the virtual clinician's office as the main menu, serving as a "virtual doorway" for emotional safety and a gentle introduction to the VR experience.

Summary

The interviews revealed both enthusiasm and caution regarding the use of VR in exposure therapy. Clinicians expressed a strong desire for realism, customization, and clinician control, while also acknowledging barriers such as cost, training, and privacy concerns. They viewed VR as most effective when used as a clinician-guided supplement to traditional treatment. These findings provided direct guidance for the tool's design, particularly the inclusion of adjustable exposure variables, a neutral onboarding space, and the possibility of clinician-led sessions. Future research and development is needed to fully address the practical constraints identified, such as equipment complexity and data security.

Phase 3: final tool development

This phase involved incorporating the insights gathered from the preceding survey and clinical therapist, and iterating on the prototype to design a final artifact. The development process was guided by the core philosophies of emotional safety, gradual challenge, and client-clinician collaboration, ensuring that the tool addressed the practical needs identified in earlier research phases (RQ1).

The final tool was developed using Unity as the game engine, providing a robust framework for VR application development. The selection of Unity as the development engine was based on its robust XR Interaction Toolkit and extensive support for cross-platform deployment. The HTC Vive was chosen as the primary target hardware due to its high-fidelity tracking and prevalence in clinical research settings (Rizzo et al., 2012). To address common SAD triggers while maintaining technical feasibility for a proof-of-concept, four environments (classroom, conference room, lecture hall, and auditorium) were selected to represent a spectrum of social density, from intimate gatherings to large-scale public speaking, allowing for the "graded exposure" central to CBT protocols.

Mirror was integrated for networking, enabling the crucial two-user (client and clinician) shared session functionality. The XR Interaction Toolkit facilitated VR input, ensuring intuitive interactions for the client. For visual assets, Blender was used for 3D model creation, and Substance Painter for texturing, with an emphasis on photorealism to enhance immersion as requested by both survey respondents and clinicians. Adobe Illustrator was employed for UI design, focusing on clarity and ease of use. An HTC Vive headset was used for VR development and testing, with one player in VR and the other in first-person mode as the clinician.

Key features were implemented based on the outcomes of Phase 1 (survey) and Phase 2 (clinician interviews).

Photorealistic 3D environment and assets

The emphasis on visual realism was a direct response to survey participants' and clinician's desire for immersion and believability to trigger genuine emotional responses. Neutral color palettes and uncluttered layouts were chosen to avoid overstimulation, aligning with emotional safety principles (Herrero et al., 2014). All 3D models in the prototype were created in Blender and textured using Substance Painter, with an emphasis on visual realism to enhance immersion. To inform this balance, visual design choices were made in consultation with clinical therapist interviews and guided by literature (Anderson et al., 2013; Herrero et al., 2014) emphasizing the importance of low-stimulation, familiar environments in anxiety treatment. For example, bright and clean lighting setups, neutral color palettes, and uncluttered layouts were used to avoid overstimulation while maintaining immersion.

Environmental props and textures were modelled after reference images of common real-world therapy and educational spaces, such as conference rooms and clinical offices. This design choice was supported by clinical therapist interviews, where participants emphasized the need for realism to evoke the right level of anxiety for exposure therapy while still ensuring the client feels safe and in control.

Clinician's office as the main menu

The VR experience begins in a 3D-modeled clinician's office, which functions as both a main menu and an onboarding space. The idea of starting the experience in a neutral, familiar environment was directly informed by clinical therapist interviews. For example, one clinical therapist stated: "There needs to be a kind of emotional doorway. Clients should not just be thrown into a scary setting" (Interviewee T1). The implementation of a 3D-modeled clinician's office was designed as an "emotional doorway" and a "safe start" environment before exposure, supporting emotional safety and gradual engagement (Craske et al., 2014).

The design of the office was based on photo references of contemporary therapy practices. The inclusion of this transitional space is supported by exposure therapy principles emphasizing predictability and gradual engagement (Craske et al., 2014). It provides a calming environment in which clients can acclimate to the virtual setting before moving into more challenging social scenarios. This onboarding phase is also important in promoting emotional regulation and helping users familiarize themselves with navigation before beginning the exposure sequence.

Scenario selection and customization user interface (UI)

A dedicated user interface (UI) panel is located within the clinician's office environment. This menu allows clinicians to configure session parameters, including the presentation setting (e.g., conference room, classroom, auditorium), audience size, audience behavior, demographic characteristics of avatars, and the format of the presentation itself. For example, clinicians can increase audience density from no audience at all to a full room, change audience attitude from passive to challenging, or choose whether the client delivers a solo speech, co-presents with the clinicians, or answers questions in a simulated Q&A format. These parameters were implemented to align with exposure therapy theory emphasizing gradual exposure, perceived control, and real-time adaptability (Powers and Emmelkamp, 2007; Herrero et al., 2014). The initial focus on a public speaking scenario was informed by survey results identifying it as a highly relevant anxiety trigger.

Multi-user and cross-play functionality

The core innovation of this tool is its support for real-time multi-user sessions, where one user (the client) enters the environment through a VR headset, while the other (the clinician) navigates the world using a standard desktop interface. Real-time collaborative multi-user sessions directly addresses the gap identified in the literature and clinician interviews regarding the lack of clinician-guided, remote VR tools. This setup enables clinician-guided sessions in both in-person and remote formats, bridging a critical gap in existing exposure therapy tools that are typically designed for solo, clinic-based use. Cross-play features enable clinicians to use the tool from a PC desktop, with or without the use of a VR headset, allowing for more flexibility.

Multi-user functionality is enabled using the Mirror networking library, which allows real-time synchronization of actions, such as scene transitions, audience changes, and slide navigation. For example, when the client changes the slide of the presentation, the change is reflected immediately in the clinician's view. While current implementation supports only two users at a time, the underlying system could be expanded to include group therapy or supervision formats in future versions.

There are some technical limitations to this approach. Latency and connection stability may affect synchronization during remote sessions, particularly if one user has significantly lower internet bandwidth. However, in local testing, synchronization remained reliable and smooth for key functions such as avatar spawning and slide transitions.

VR and input integration

The tool was developed using the XR Interaction Toolkit, supporting both the HTC Vive headset for immersive VR and the Unity XR Simulator for desktop testing. This allows the tool to accommodate both users with full VR setups and those without headsets, broadening accessibility.

Interaction in VR is based on laser pointer selection through hand controllers, allowing clients to interact with UI panels, navigate menus, and toggle between settings. For clinicians, traditional mouse and keyboard controls are available, supporting familiarity and reducing the learning curve for clinicians less experienced with VR systems. Support for both HTC Vive and Unity XR Simulator was a pragmatic decision to broaden accessibility, acknowledging the cost barriers identified by clinicians.

Audience spawn system

To simulate the core component of social exposure (being observed by others) the prototype includes a dynamic audience

spawning system (i.e., automatically generating a digital audience). The dynamic audience spawning system with adjustable density settings directly supports the principle of progressive exposure (Chowdhury and Khandoker, 2023) and responds to clinician requests for customizable exposure levels. The clinician can adjust the number of audience avatars present in the room using the UI panel. Available density settings include no audience (0% of seats filled), small (20%), medium (50%), and large (100%). These increments allow clinicians to introduce clients gradually to different levels of social pressure, consistent with the principle of progressive exposure (Chowdhury and Khandoker, 2023).

This development phase served as the tangible output of the design process, embodying the accumulated knowledge and feedback from the preceding research phases. This phase focused on developing a minimum viable product (MVP) to test core functionalities and design principles. At this point in development, only one primary scenario (public speaking) was implemented; however, results from the survey and interviews showed interest in scenarios like job interviews, which will be updated in future versions of the tool. The underlying architecture was explicitly designed to support modular expansion, to support generalization and future feature integration.

Phase 4: playtesting

Playtesting served as the final evaluative phase within the current study, providing direct user feedback on the prototype's usability, clarity, and perceived therapeutic usefulness. This empirical data was vital for assessing the effectiveness of the design decisions made in earlier phases and for identifying areas for future refinement.

Participants

Participants for the playtesting were recruited from the initial survey, in which respondents were given the option to express interest in trying the VR prototype. A total of eleven individuals, out of 16 contacted who expressed interest, participated in the testing phase. Of these, eight were aged between 18 and 24 years, while the remaining three were within the 25–34 age group (M = 23). All participants took on the role of the client and interacted with the prototype using the Unity XR simulator. No participants played the role of the clinician during testing, as the primary aim was to assess the client-side experience of usability and clarity for this initial prototype evaluation (clinician experiences were previously assessed in the interview phase). This pragmatic choice allowed for broader accessibility in testing and focused the evaluation on core user interactions.

Materials and procedure

The Unity XR simulator was used to emulate VR interactions through mouse and keyboard input, enabling users to navigate menus, interact with the environment, and operate scenario functions like slide advancement. This way, participants did not require a VR headset and could interact with the prototype via a regular computer screen. Each session began with the participant entering a 3D-modeled virtual clinician's office, which served as the onboarding environment. They were then guided through the customization of the public speaking scenario, including selection of the environment type and audience size. Once inside the presentation space, they were instructed to interact with key features such as navigating PowerPoint slides using virtual controls. In addition to the core interactions, participants were encouraged to explore the virtual world freely to assess the interface's intuitiveness and environmental clarity.

This choice was further motivated by the need to isolate the software's instructional and therapeutic clarity from the common confounds of immersive environments. Specifically, full immersion can introduce significant confounding variables such as the novelty effect of VR, which may cause participants to overlook usability flaws due to the initial sensory impact, as well as physical discomfort associated with cybersickness. By utilizing a 2D simulator, we ensured that participant feedback was focused strictly on the application's information architecture and the perceived relevance of the therapeutic scenarios, providing a cleaner assessment of the tool's functional logic before moving to full immersive deployment.

After the session, each participant completed a digital post-test questionnaire consisting of 15 items. These items used a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) to assess three main areas: usability and navigation, scenario clarity and relevance, and perceived therapeutic usefulness. The questionnaire was designed to gather quantitative feedback on key aspects of the user experience, complementing the qualitative design insights gained from clinical therapist interviews and survey responses. Testing sessions lasted approximately 15–20 min, depending on the participant's pace and engagement with the environment.

Data analysis

All survey responses (n = 11) were compiled and analyzed. For each of the 15 items, the mean value and standard deviation were calculated to understand the central tendencies and variability of user responses. These quantitative results were used to evaluate the overall success of the prototype in meeting design goals related to usability, clarity, and therapeutic alignment.

Results

The playtesting results indicated strong positive reception for the prototype, while also highlighting areas for refinement. See Table 1 for a summary of the results.

TABLE 1 Mean ratings and standard deviations for prototype evaluation.

Category	Item	M	SD
Usability	Navigation ease	4.55	0.89
Relevance	Scenario purpose	4.64	0.64
Efficacy	Remote potential	4.55	0.6

Usability and navigation

Results showed that users found basic interactions straightforward. Overall menu and navigation ease of use

received high ratings (M = 4.55; SD = 0.89), as well as presentation controls (M = 4.73; SD = 0.45), suggesting successful core interaction design. These high ratings suggest that the fundamental interaction design and information architecture of the prototype are robust, providing a solid foundation for future development.

Scenario clarity and relevance (RQ3)

Participants gave high ratings for understanding the scenario's purpose (M = 4.64; SD = 0.64) and its relevance for individuals with social anxiety (M = 4.64; SD = 0.48), validating the chosen public speaking scenario. The item "The level of exposure in the scene felt appropriate" scored slightly lower (M = 4.09; SD = 0.83), indicating an expected broader range of user perceptions regarding challenge. This variation in perceived exposure levels suggests a need for further refinement in future iterations to better suit a range of user sensitivities and ensure optimal therapeutic challenge.

Effectiveness and future improvements (RQ3)

Participants found the prototype potentially helpful for individuals who avoid in-person therapy (M = 4.55; SD = 0.66) and expressed interest in more customization options (M = 4.55; SD = 0.66). The statement "The prototype gave me an idea of what a real exposure therapy session might look like" received a mean score of 4.27 (SD = 0.63). A participant also noted that "NPCs need more detail to fit in with the rest of the graphics," suggesting visual polish for avatars could enhance immersion. Enhancing NPC realism is an identified area for future improvement to further enhance immersion and the overall therapeutic impact.

Design implications

The playtesting outcomes confirmed the prototype's foundational success and provided clear directions for future iterations, directly influencing refinements to address RQ3 (user experience).

The high perceived usefulness for people who avoid in-person therapy reinforces the potential for the tool to support blended care models. It is important to incorporate on seamless integration with existing telehealth platforms and therapy management systems, making it easy for clinicians to incorporate the *SafeSpaceVR* tool into their current practice. Additionally, results highlighted the importance of the dynamic scaling of exposure levels (e.g., granular control for clinicians, adaptive difficulty based on real-time user input).

The generally positive user experience findings (high usability, clarity, and perceived usefulness) in this initial evaluation strongly suggest the prototype's readiness for more rigorous clinical validation. This implies the next steps should involve conducting formal studies with actual patients undergoing therapy for social anxiety, to assess real-world efficacy, engagement, and long-term therapeutic outcomes.

Discussion

The current study explored how a clinician-guided virtual reality tool could be designed to support exposure therapy for social anxiety, while also offering a generalizable model for immersive clinical tools. We used a four-phase process (surveys, interviews, final development, and playtesting) within a UCD framework to design and test the *SafeSpaceVR* tool for use by clinicians. The resulting tool is a functional, multi-user VR application developed in Unity, designed and tested to support clinicians in delivering customizable scenarios for therapeutic treatments, particularly exposure-based therapy for social anxiety. While the design and evaluation of this tool focused on social anxiety use cases, its technical and interaction framework allows for broader applications across multiple therapeutic modalities, including talk therapy, cognitive-behavioral interventions, and phobia desensitization.

Through a combination of clinical therapist interviews, user surveys, and playtesting, several key insights emerged. Survey results helped identify the most socially stressful scenarios for users, namely public speaking, phone calls, and interacting with strangers, which directly shaped the selection of the first prototype scenario. Interviews reinforced the clinical relevance of these use cases and provided valuable suggestions that guided both the tool's structure and its clinician-facing features. Clinicians also highlighted the importance of realism, emotional safety, and control over exposure settings. In response, the prototype incorporated customizable environments, a transitional onboarding space, and clinician-side controls, which received high usability ratings in user testing. Playtesting further demonstrated that the tool was intuitive, well-received, and effective in simulating a therapeutic experience.

Alignment with theoretical frameworks

The design of the prototype aligns closely with the two guiding theoretical frameworks: exposure therapy principles and player-centered design with emotional safety. In accordance with exposure therapy theory, features like adjustable audience sizes, real-time clinician presence, and a neutral onboarding space allow for graded exposure–a core therapeutic method shown to promote habituation and reduce avoidance (Craske et al., 2014; Powers and Emmelkamp, 2007). This structure enables the clinician to gradually introduce stressors in a controlled way, helping clients to confront anxiety-provoking situations at their own pace.

Just as important, the design integrates principles of player-centered interaction and emotional safety. The use of a calm, familiar clinician's office as the starting point reflects design practices aimed at easing users into more emotionally intense experiences (Herrero et al., 2014). This onboarding environment functioned as a "virtual doorway," as one clinician phrased it, giving users a chance to acclimate before entering a socially challenging scenario. This suggests that emotionally sensitive VR design must include more than just scenario content; it must also carefully consider user onboarding, pacing, and perceived safety.

During the development phase, a key design tension emerged between visual realism and functional utility. While participants initially expressed a desire for high-fidelity "photorealistic" avatars, clinician feedback emphasized that "clear behavioral cues" (e.g., eye contact, nodding, or whispering) were more therapeutically relevant than graphical resolution. Consequently, we prioritized the development of an interaction-heavy architecture over visual complexity. This insight suggests that for VRET, psychological

fidelity, or the degree to which the virtual agents behave in socially relevant ways, is a more critical design requirement than "graphic fidelity."

Clinician perspectives and clinical alignment

Clinicians in this study reported the potential of the *SafeSpaceVR* tool to be integrated into real clinical workflows. They emphasized the importance of maintaining a clinical presence, customizing scenarios, and monitoring client behavior, needs that current solo-use platforms like Psious or Amelia Virtual Care often fail to meet (Freeman et al., 2017; Meyerbröker, 2021).

This feedback not only informed specific features but also shaped the conceptual design: rather than replacing therapy, the *SafeSpaceVR* tool serves as a supplement that empowers clinicians and enhances client engagement. This demonstrates that co-designing tools with clinicians may be essential for meaningful clinical integration.

A significant insight derived from the UCD process was the necessity of functional asymmetry. While previous VRET models often sought to immerse both the clinician and patient, our findings suggest that immersion for the clinician can introduce a 'cognitive bottleneck.' By providing the clinician with a high-level 2D dashboard while the patient is immersed, *SafeSpaceVR* preserves the clinician's role as an objective observer and scaffold. This allows the therapist to maintain situational awareness, consult clinical notes, and manage the technical environment without the sensory load of VR, thereby strengthening the therapeutic alliance through more focused, real-time responsiveness.

Implications for mental health practice

This study suggests that clinician-guided VR applications can meaningfully enhance traditional and remote therapy models. A particular innovation of the *SafeSpaceVR* tool designed in this study is the inclusion of multi-user functionality and clinician-side controls. These features not only strengthen the broad clinical application of the tool, but also suggest that future mental health technologies may also have the potential to go beyond patient-facing design and more deeply integrate clinician workflows and preferences into digital tools. The ability to customize exposure tasks, observe client behavior in real time, and gradually build challenge levels replicates the core mechanisms of exposure therapy. Moreover, clinicians indicated that such tools could provide controlled scenarios when *in vivo* options are unavailable, such as for clients with limited social contact or mobility constraints.

The tool also aligns with blended care models, where digital tools support but do not replace therapy. The architectural design of *SafeSpaceVR* specifically facilitates a blended care model, wherein digital interventions serve as an augmentative scaffolding rather than a replacement for the therapist-client relationship. Our findings suggest that the tool can occupy a unique position in the hierarchy of exposure. Traditionally, patients move from imaginal exposure (in-office) to *in vivo* exposure (real world), but the transition is often steep and poorly monitored. *SafeSpaceVR* acts as a mediating environment: it offers higher ecological validity than imagination while maintaining a level of safety and clinician-mediated control that the real world lacks.

Furthermore, the participants in this study highlighted the potential for hybrid deployment. In this model, the initial "heavy lifting" of exposure is conducted in live, clinician-guided sessions (asymmetric mode), where the therapist can calibrate the environment to ensure the patient does not become overwhelmed. Once a baseline of safety is established, the same virtual environments can be assigned as autonomous homework modules. This allows for the repetition and frequency required for habituation and inhibitory learning, while the open-source nature of the platform ensures that patients can access these environments remotely without the burden of proprietary licensing fees. This approach aligns with the growing shift toward digitally-augmented CBT, where the clinician focuses on high-level cognitive restructuring while the VR tool provides the standardized, repeatable stimulus for exposure (Lau et al., 2024).

Addressing structural barriers through open-source design

A central contribution of this study is the development of *SafeSpaceVR* as an open-source research artifact. During the clinician interviews (Phase 2), participants identified significant structural barriers to VR adoption, including high equipment costs, a lack of specialized therapist-centered tools, and limited technical training. By utilizing a Research through Design (RtD) methodology, we have created a prototype that is not only shaped by clinical workflows but is also transparent and extensible.

Our design process highlighted a significant frustration among clinicians regarding the "black box" nature of proprietary VRET tools. When clinicians cannot see the underlying logic or modify the environment to suit a specific patient's trauma or trigger, the tool becomes a barrier rather than an aid. By providing SafeSpaceVR as an open-source artifact, we contribute to the "democratization of clinical VR." This allows institutions to audit the code for privacy, researchers to add custom environments (e.g., culturally specific social settings), and practitioners to avoid the prohibitive licensing costs that currently gate-keep VRET from lower-resource clinical settings.

Unlike proprietary "black box" software, the open-source nature of *SafeSpaceVR* directly addresses the financial and accessibility barriers cited by practitioners. By providing the Windows executable (.exe) and making the design architecture public, this project lowers the barrier to entry for clinics with limited budgets and allows for community-driven customization. Furthermore, as a proof of concept, *SafeSpaceVR* demonstrates the feasibility of developing low-cost, high-utility tools that prioritize the clinician's role as an active guide within the virtual space, rather than a passive observer outside of it. This shift from patient-only platforms to clinician-guided environments represents a critical step in making immersive therapy both practically viable and clinically effective in diverse settings.

Broader applications beyond social anxiety

Although this study focused on social anxiety, the design architecture has broader relevance. The modular scenario system

could support exposure exercises for other conditions, such as phobias (e.g., fear of flying), trauma related disorders (e.g., revisiting difficult situations), or even speech rehabilitation contexts. The results here add to the growing body of studies with VR that show its utility in addressing psychological phenomenon (Garcia-Palacios et al., 2001; Juvrud et al., 2018; Powers and Emmelkamp, 2007; Zhang and Juvrud, 2024). One of the unique contributions of the current study is the clinician-controlled and multi-user format, which has potential value in talk therapy or psychoeducation, where co-presence and environmental context may support empathy or skill practice. Additionally, the tool could be used as a training platform for future clinicians, allowing them to practice (exposure) therapy exercises in their education. As such, this could be a valuable tool for schools, universities, and training programs.

Future directions

There are several areas of planned future research and refinement, directly addressing user experience, usability, clarity, and perceived usefulness for therapeutic purposes (RQ3). First, the tool can be expanded to include additional scenario types, such as job interviews, social gatherings, or customer service interactions. These were frequently requested in both survey responses and clinical therapist interviews and would significantly broaden the tool's clinical relevance and perceived usefulness for a wider range of social anxiety triggers.

Features that facilitate more comprehensive clinician monitoring should also be prioritized. This includes functionalities such as live camera views, session recording, and objective data tracking. While the current prototype allows clinicians to control and observe scenes, advanced metrics like gaze tracking or physiological data were of significant interest to clinicians and would provide richer insights into client responses, thereby enhancing the perceived usefulness and clinical utility of the tool.

Tracking therapeutic progress is another area where VR offers unique potential. Traditionally, progress is measured through self-reports and clinician observations. Heimberg et al. (2010) introduced the Social Anxiety Session Change Index (SASCI), which offers a more structured, quantitative way to monitor improvements. In VR, physiological metrics such as heart rate or gaze tracking can offer real-time, objective feedback, a feature supported in principle by research from Parsons and Rizzo (2007). Several clinical therapists interviewed during this project expressed interest in such features, particularly for supplementing subjective session notes.

Rigorous clinical validation is essential

Future studies should assess therapeutic outcomes by utilizing the tool in real therapy sessions and case studies with clients undergoing treatment for social anxiety or related conditions. This will enable a direct comparison of VR exposure therapy with traditional methods in terms of efficacy, engagement, and dropout rates, further validating its perceived usefulness and effectiveness.

Finally, for successful integration into mental healthcare, comprehensive clinician training and robust user onboarding materials are necessary, particularly for clinicians with lower digital literacy. Clinicians must feel confident and competent in using the tool, and patients need clear expectations and guidance. Instructional materials, walkthroughs, and guided demonstrations will be vital components in any full deployment, directly enhancing the clarity and usability for both clinicians and clients.

By addressing these future directions, this VR exposure therapy tool has the potential to evolve into a highly flexible and impactful resource within the rapidly changing landscape of digital mental healthcare. The foundation laid in this project provides a solid base upon which more complex, inclusive, and scalable VR therapy solutions can be built.

Conclusion

This paper set out to investigate how a VR exposure therapy tool could be designed to meet the needs of clinicians and patients in treating SAD, particularly within blended or remote therapy contexts. The research was guided by three core questions: How to design a VR tool that aligns with therapeutic needs (RQ1), which scenarios are most relevant for VR-based exposure (RQ2), and how users perceive the tool in terms of usability and clinical potential (RQ3).

The final *SafeSpaceVR* tool enables real-time interaction between clinicians and client in a shared virtual space, offering customizable environments that simulate public speaking tasks. Both quantitative and qualitative data suggest strong user interest and clinician support for such tools. The survey revealed that the most anxiety-provoking social situations, such as public speaking and job interviews, are also the ones users would most like to practice in VR. Interviews with clinicians emphasized the importance of emotional safety, realism, and clinician control. Playtesting further confirmed that the tool is usable and relevant, with high ratings for clarity and engagement. However, as a small-scale proof of concept, these findings should be viewed as an evaluative baseline. The positive reception regarding usability and clarity suggests that *SafeSpaceVR* is a viable candidate for further clinical validation studies, which must include rigorous testing by mental health professionals in live therapeutic contexts to fully establish its place within clinical workflows.

While social anxiety served as the initial focus, the underlying design model (real-time clinician-guided scenarios with adjustable levels of challenge) could be extended to support a range of disorders and treatment formats. Furthermore, this model may offer value in psychoeducation, trauma processing, communication skills training, or even talk therapy when paired with shared virtual environments. The ability to co-regulate experiences, customize stimuli, and repeat difficult interactions in safe, immersive spaces opens new possibilities for therapeutic engagement. As remote and hybrid therapy become increasingly normalized, especially post-pandemic, VR tools can play a key role in maintaining therapeutic presence, even when clinicians and client are not physically co-located.

From a technical standpoint, the choice of the Mirror networking library ensures that SafeSpaceVR is not limited to one-on-one sessions. While the current proof of concept focuses on a dyadic interaction, the underlying architecture is architecturally scalable; it can support larger concurrent user groups, potentially accommodating group therapy

sessions or clinical supervision formats with 10–50 synchronized avatars, without significant latency, provided sufficient server bandwidth. This scalability positions the tool as a robust foundation for evolving clinical needs, from individual practice to institutional training environments.

Taken together, the findings demonstrate a preliminary but promising step that clinician-guided VR applications can serve as viable clinical tools for patients when carefully designed around emotional safety, gradual exposure, and collaborative flexibility. Although the current version focuses on social anxiety, its architecture and interaction model are flexible enough to support other psychological disorders and therapy formats. This initial feasibility study demonstrates strong patient-side usability and relevance, but we acknowledge that the full-cycle validation of SafeSpaceVR within a live clinical setting, incorporating real-time therapist feedback during active treatment, is the necessary next step to confirm its alignment with professional clinical workflows. Nonetheless, by framing the system as a generalizable platform for clinician-led interventions, this research contributes to wider discussions on digital health equity, remote care delivery, and the ethical implementation of immersive technologies in clinical settings. As a result, this tool contributes not only a working VR application but a roadmap for future tool development that balances immersion with clinical realism, technical feasibility with emotional safety, and innovation with therapeutic ethics.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by Swedish Ethical Review Authority (2023-04552-02). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

KH: Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. JJ: Formal Analysis, Supervision, Project administration, Writing – review and editing.

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